

behind it have a responsibility to not ignore those who can truly make the most use from it.

Thankfully, the branch of Assistive Technology focuses on doing just that. A variety of products and services are now available in the market for the disabled, helping them live near ordinary lives.

A rainbow for the unsighted

Less than 5 per cent of India's population has access to education as of today. Despite the claim of 'education-for-all', matters such as these are treated as welfare issues rather than government policies. Even as India turns out to be one of the superpowers in the world with rapid developments in technology, resources and manpower, a sluggish execution and lack of support from the Central government nulls whatever efforts are made for the disabled. It is only through private institutions and NGOs that technology stands a better chance of penetration to these masses.

With their sense of vision either gone or severely impaired, those suffering from any form of vision loss must make more use of their other senses in order to communicate. Aside from spoken words, Braille is a most effective way for the visually impaired to 'read'. As a majority of the blind population is familiar with Braille, it is a very useful medium for education.

Accessing personal computers is made much easier with the use of Braille peripherals. Though technologically inferior to modern methods of sharing information with the blind, Braille-based readers work by raising and lowering a different combination of pins electronically, to represent in Braille whatever content appears on the screen. Braille printers do the actual embossing of Braille onto paper. These are the counterparts of inkjet printers but instead of ink, special pins are used to create a physical impression on paper. Of all PC-related Braille devices, none may be as essential for a blind person as Electronic Braille Note takers. These are small, portable devices used for recording information entered through the use of a Braille keyboard. For output, note takers use a Braille display for speech synthesis for spoken voice. One of the more popular electronic note taking devices comes from Freedom Scientific and is called Braille 'n Speak. It features a word proces-

sor complete with spell checking capabilities. Advanced versions of a note taker are in the form of fully featured notebook computers complete with Braille display and synthetic speech. Like a conventional laptop, mobile PCs for the blind come with CD-ROM drives, a generous helping of memory, and multimedia sound. The only difference lies in the means of actually reading content.

Technology for the blind may have reached greater heights with the creation of the Dobelle Eye, the world's first artificial eye. By attaching a camera directly wired to the brain and mounted to a pair of sunglasses, partial vision was restored in the first recipient of this technology. Known as Jeff, the 62-year-old has been blind since age 36. With the Dobelle Eye, Jeff can now read letters that are 2 inches tall from 5 feet away. An ultra-

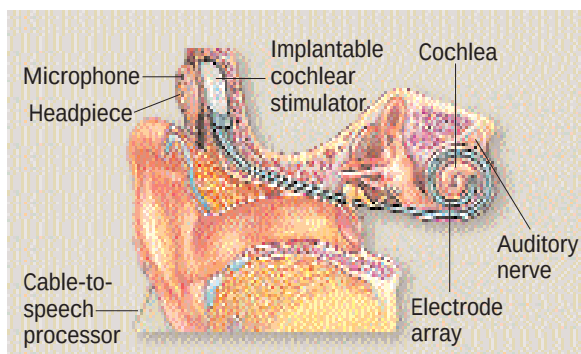
sonic range finder makes up the second lens of the pair of sunglasses. Both the finder and lens communicate with each other using a computer system carried on Jeff's waist. The electrodes attached to the brain stimulate brain cells allowing the user to see instances of light and limited vision.

This technology is still in its infancy and though Jeff can see enough to put a cap on the head of a mannequin, the totality of vision is far less than what a typical sighted man is used to.

A note for the deaf

A simple telephone conversation, which we take so much for granted, is out of reach for those with hearing losses. TTY, which stands for Telecommunication Typewriter, is a relay service that enables a deaf person to type in messages that are read out by an intermediary operator to the recipient. It enables the deaf to communicate, but it's not without problems. TTY is a bit slow when it comes to communicating messages. The service can often be unreliable and a typical 2 minute conversation extends to 20 minutes with the use of TTY.

With a cure for deafness many years away, medical researchers pushed technology to the next best thing, a cochlear implant. A cochlear implant is a tiny electronic piece fitted within the skull just behind the ear. This type of implant does not give the user natural sounds; the sound is a bit synthesised but it restores some of the ability to hear.



Technology has helped restore hearing with this device by translating sound energy into nerve impulses and sending those impulses to the brain

Biochips

If you were asked to inject a microchip into your head, would it sound like a bad case of X-Files obsession? Well, the truth, as they say, is out there. Researchers are looking to go skin deep and use chip implant technology for human health reasons. Such a chip could be injected under the skin of diabetics to monitor glucose levels and send the results out through radio-frequency communication.

Cardiac pacemakers have long held up the lives of those with heart ailments. Though the device works, it is far from perfect. In an attempt to circumvent complications associated with the age-old pacemaker, biochips can be used to regulate the process of abnormal heart rhythms after being inserted into the brain.

While the best treatment available for hearing loss comes in the form of a cochlear implant, the next area of focus would be on vision. Groups of researchers are in the process of developing a chip implant for the eye where the chip would mimic the various functions involved in sight. The duo of Joseph Rizzo from the Massachusetts Eye and Ear Infirmary and John Wyatt of the Massachusetts Institute of Technology, co-directors of the Retinal Implant Project, are currently working on an aid for the visually impaired based upon Biochip technology. Situated on a pair of glasses, a specialised Charged-Couple Device (CCD) captures and digitises images of the outside world, encodes it, and uses a signal to stimulate healthy neural cells.